Positive Factorization Through Free Banach Lattices
Verification, Literature Status, and Further Reductions

Research report prepared from the supplied partial solution

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Abstract

Let X be a Banach lattice, let $\delta_X : X \to \mathrm{FBL}[X]$ be the canonical linear isometry, and let $\beta_X : \mathrm{FBL}[X] \to X$ be the canonical contractive lattice homomorphism satisfying $\beta_X \delta_X = \mathrm{Id}_X$. Oikhberg asked whether β_X always has a positive bounded linear right inverse.

The supplied argument is correct: it gives an explicit positive section for $\ell_1(I)$ and transfers it to every positive retract of $\ell_1(I)$. We also identify the exact scope of that hypothesis: such retracts are precisely, up to lattice isomorphism, the spaces $\ell_1(A)$. A literature search through 21 June 2026 found no published or preprint proof or counterexample for the general question. The report records several additional affirmative results and reductions. In particular, Lotz's positive lifting theorem implies the answer is affirmative for every AM-space, hence for every $C_0(K)$ and every $c_0(\Gamma)$. We prove an equivalent split-quotient formulation, construct a canonical positive contractive section into $\mathrm{FBL}[X]^{**}$ for every X, and show that the remaining obstruction is precisely weak-star continuity of a positive dual retraction. Finally, we analyze the natural construction for nonatomic L_1 and show concretely why it leaves the free Banach lattice and lands only in its bidual. The arbitrary Banach-lattice problem remains open.

1 The problem and the conclusion of this investigation

For a Banach space E, the free Banach lattice $\mathrm{FBL}[E]$ comes with a canonical linear isometry $\delta_E : E \to \mathrm{FBL}[E]$. Every bounded operator $T : E \to Y$ into a Banach lattice has a unique lattice-homomorphic extension

$$\widehat{T} : \mathrm{FBL}[E] \longrightarrow Y, \qquad \widehat{T}\delta_E = T, \qquad \left\|\widehat{T}\right\| = \|T\|.$$

If X itself is a Banach lattice, applying this property to Id_X gives a contractive lattice homomorphism

$$\beta_X : \mathrm{FBL}[X] \longrightarrow X, \qquad \beta_X \delta_X = \mathrm{Id}_X .$$

Thus δ_X is always a bounded linear section of β_X, but it is usually not positive. The question raised in [8, Section 3] is:

Question 1.1 (Oikhberg)**.** *Does every Banach lattice X admit a positive bounded linear map $U : X \to \mathrm{FBL}[X]$ such that $\beta_X U = \mathrm{Id}_X$?*

Our literature search included the exact wording of the question, the phrases "positive factorization through $\mathrm{FBL}[X]$", "positive right inverse", and "split lattice quotient", citations to [8], and recent work on free, projective, semiprojective, and pre-ordered Banach lattices. We found no paper resolving Question 1.1. In particular, the recent construction of free Banach lattices over pre-ordered Banach spaces in [4] uses a different universal object whose generating map is positive by definition; it does not produce a positive section into the ordinary FBL[X]. Recent work on complemented subspaces and free products likewise does not settle the question [3, 6]. A literature search cannot exclude unpublished work, but the best supported status as of the date of this report is that the general problem remains open.

The main concrete conclusions below are:

(i) the supplied partial proof is valid;

(ii) its positive-ℓ_1-retract hypothesis is equivalent to being lattice isomorphic to an atomic L_1-space $\ell_1(A)$;

(iii) all AM-spaces have the desired positive factorization, by a classical theorem of Lotz;

(iv) the problem is equivalent to asking whether every linearly split lattice quotient onto X splits positively;

(v) every X has a canonical positive contractive section into FBL[X]**, and the exact remaining issue is whether a suitable dual retraction can be chosen weak-star continuous.

2 Verification of the supplied partial result

We first reproduce the argument in a slightly sharpened form. The proof is correct, including for arbitrary index sets.

Lemma 2.1 (The atomic ℓ_1 lifting)**.** *For every set I there is a positive contraction*

$$S_I : \ell_1(I) \longrightarrow \mathrm{FBL}[\ell_1(I)]$$

such that $\beta_{\ell_1(I)} S_I = \mathrm{Id}_{\ell_1(I)}$.

Proof. Let $(e_i)_{i\in I}$ be the canonical atoms of $\ell_1(I)$. For $a = (a_i)_{i\in I} \in \ell_1(I)$ put

$$S_I a = \sum_{i\in I} a_i\, \delta_{\ell_1(I)}(e_i)^+.$$

The net of finite partial sums converges in norm, since

$$\sum_{i\in I} |a_i| \left\| \delta_{\ell_1(I)}(e_i)^+ \right\| \le \sum_{i\in I} |a_i| < \infty.$$

Equivalently, every element of $\ell_1(I)$ has countable support and the resulting series is absolutely convergent. Hence S_I is a contraction. It is positive because $a_i \ge 0$ for all i implies $S_I a \ge 0$. Finally, $\beta_{\ell_1(I)}$ is a lattice homomorphism, so

$$\beta_{\ell_1(I)}(\delta(e_i)^+) = \beta_{\ell_1(I)}(\delta(e_i))^+ = e_i.$$

Linearity and norm convergence now give $\beta_{\ell_1(I)} S_I a = a$. □

Theorem 2.2 (The supplied positive-retract theorem)**.** *Suppose there are positive bounded maps*

$$J : X \longrightarrow \ell_1(I), \qquad Q : \ell_1(I) \longrightarrow X, \qquad QJ = \mathrm{Id}_X .$$

Then there is a positive map $U : X \to \mathrm{FBL}[X]$ satisfying $\beta_X U = \mathrm{Id}_X$ and

$$\|U\| \le \|Q\|\, \|J\| .$$

Proof. Let S_I be as in Theorem 2.1 and let $\mathrm{FBL}[Q] : \mathrm{FBL}[\ell_1(I)] \to \mathrm{FBL}[X]$ be the lattice homomorphism induced by the bounded linear map Q. Define

$$U = \mathrm{FBL}[Q] S_I J.$$

This is positive and has the stated norm bound. If $Jx = (a_i)$, then

$$\begin{aligned}
\beta_X U x &= \sum_i a_i\, \beta_X\, \mathrm{FBL}[Q]\,(\delta(e_i)^+) \\
&= \sum_i a_i\, \beta_X\,(\delta_X(Qe_i)^+) \\
&= \sum_i a_i Q e_i = QJx = x.
\end{aligned}$$

The third equality uses $Qe_i \geq 0$. All series are absolutely convergent in norm. $\square$

Remark 2.3 (Two points worth recording)**.** *First, the last calculation is genuinely needed. The identity*

$$\beta_X\, \mathrm{FBL}[T] = T \beta_Y$$

is automatic when $T : Y \to X$ is a lattice homomorphism, because both sides are then lattice homomorphisms agreeing on the free generators. It need not hold for a merely positive T. Thus the positivity of each Qe_i is the key special feature of the atomic proof.

Second, the finite-dimensional corollary is correct but was already known in a stronger form: finite-dimensional Banach lattices have the lattice-lifting property, in which the section itself can be chosen to be a lattice homomorphism [2, Proposition 2.2].

3 What the ℓ_1-retract hypothesis really means

The partial theorem appears at first to cover a broad class of positive retracts. In fact, that class is exactly the atomic L_1 class.

Theorem 3.1 (Classification of positive ℓ_1 retracts)**.** *For a Banach lattice X, the following are equivalent.*

- *(a) There are a set I and positive bounded maps $J : X \to \ell_1(I)$ and $Q : \ell_1(I) \to X$ with $QJ = \mathrm{Id}_X$.*
- *(b) X is lattice isomorphic to $\ell_1(A)$ for some set A.*

Proof. Only (a)⇒(b) needs proof. Let $\mathbf{1} = (1)_{i \in I} \in \ell_\infty(I)$ and define the positive functional

$$\varphi = J^* \mathbf{1} \in X_+^*.$$

Set

$$|||x||| = \varphi(|x|) = \| J|x| \|_{\ell_1(I)}\,.$$

This is an equivalent lattice norm. Indeed,

$$|||x||| \leq \|J\|\, \|x\|\,,$$

while positivity of J gives $|Jx| \leq J|x|$, and hence

$$\|x\| = \|QJx\| \leq \|Q\|\, \|Jx\|_1 \leq \|Q\|\, |||x|||.$$

For $x, y \geq 0$,

$$|||x + y||| = \varphi(x + y) = |||x||| + |||y|||.$$

Thus X, with this equivalent lattice norm, is an abstract AL-space. By the Kakutani representation theorem it is lattice isometric to an $L_1(\mu)$ space.

On the other hand, J is an isomorphic embedding of the underlying Banach space into $\ell_1(I)$, so X has the Schur property. The Schur property is unchanged by equivalent norms. An $L_1(\mu)$ space has the Schur property only when its measure algebra is purely atomic: a nonzero nonatomic finite part supports a norm-one weakly null Rademacher sequence. A purely atomic L_1 space is lattice isometric, after absorbing the atom weights, to $\ell_1(A)$ for a suitable set A. This proves (b). The converse follows by taking $I = A$ and the identity maps. $\square$

Corollary 3.2. *The new content of Theorem 2.2, up to lattice isomorphism, is precisely the positive factorization for the spaces $\ell_1(A)$.*

This also corrects one limitation stated in the supplied packet. The packet said that it gave no answer for c_0. Although c_0 is not in the class of Theorem 3.1, it does have the desired factorization, and even the stronger lattice-lifting property; see the next section.

4 Affirmative classes already available in the literature

4.1 The stronger lattice-lifting property

Avilés, Martínez-Cervantes, Rodríguez, and Tradacete call X a lattice lifting space if there is a lattice homomorphism $A : X \to \mathrm{FBL}[X]$ with $\beta_X A = \mathrm{Id}_X$ [2]. This is stronger than Question 1.1. Their results include the following.

- Every projective Banach lattice has the lattice-lifting property; in particular this includes finite-dimensional Banach lattices, ℓ_1, and $C[0, 1]$.
- Every free Banach lattice $\mathrm{FBL}[E]$ has the isometric lattice-lifting property.
- Every Banach space with a 1-unconditional Schauder basis, equipped with its coordinate order, has the isometric lattice-lifting property. Consequently, the classical separable spaces c_0 and ℓ_p have it.

The same paper shows that the stronger property fails for nonatomic L_p spaces and for $c_0(\Gamma)$ and $\ell_p(\Gamma)$ when Γ is uncountable. Those negative results do not answer Question 1.1, since a positive section need not preserve disjointness.

4.2 A classical theorem of Lotz gives all AM-spaces

The following consequence of a 1975 theorem of Lotz appears not to have been noted in the discussion surrounding Question 1.1.

Theorem 4.1 (Lotz)**.** *Every AM-space X admits a positive bounded map $U : X \to \mathrm{FBL}[X]$ such that $\beta_X U = \mathrm{Id}_X$. Consequently the same is true for every Banach lattice lattice-isomorphic to an AM-space.*

Proof. Lotz proved that every AM-space is *d-projective*: if $q : Y \to Z$ is a nearly interval-preserving metric surjection of Banach lattices and $T : X \to Z$ is positive, then T has a positive bounded lifting through q [5, Proposition 3.3].

We verify the hypotheses for $q = \beta_X$. It is a metric surjection: for $x \in X$,

$$\inf\{\|f\| : \beta_X f = x\} = \|x\|,$$

because β_X is contractive and $\delta_X x$ is a preimage of norm $\|x\|$. It is in fact interval preserving. If $f \geq 0$ in $\mathrm{FBL}[X]$ and $0 \leq x \leq \beta_X f$, choose $g \in \mathrm{FBL}[X]$ with $\beta_X g = x$ and put $h = f \wedge g^+$. Then $0 \leq h \leq f$ and, since β_X is a lattice homomorphism,

$$\beta_X h = \beta_X f \wedge (\beta_X g)^+ = \beta_X f \wedge x = x.$$

Apply Lotz's theorem to the positive map $T = \mathrm{Id}_X : X \to X$. □

Corollary 4.2. *The answer to Question 1.1 is affirmative for every $C_0(K)$ space and every $c_0(\Gamma)$, with arbitrary locally compact K and arbitrary set Γ. In particular, uncountable $c_0(\Gamma)$ has a positive section although it has no lattice-homomorphic section.*

For orientation, the present state of several standard examples is summarized below. "Lattice section" means the stronger property from [2].

Space or class	Positive section	Lattice section
$\ell_1(\Gamma)$	yes, explicit	no in general if Γ uncountable
$c_0(\Gamma)$	yes, Lotz	no in general if Γ uncountable
$C_0(K)$ / AM-spaces	yes, Lotz	not in general
1-unconditional Schauder basis	yes	yes
$\mathrm{FBL}[E]$	yes	yes, isometric
finite-dimensional lattices	yes	yes
nonatomic L_1	open	no

5 An exact split-quotient reformulation

The next proposition turns the free-lattice question into a problem purely about split lattice quotients. It is the positive analogue of [2, Proposition 2.1].

Define

$$\mathrm{pf}(X) = \inf\{\|U\| : U : X \to \mathrm{FBL}[X] \text{ is positive and } \beta_X U = \mathrm{Id}_X\},$$

with $\mathrm{pf}(X) = \infty$ if no such map exists.

Proposition 5.1 (Split-quotient characterization)**.** *Let X be a Banach lattice and $\lambda \geq 1$. The following are equivalent.*

(a) $\mathrm{pf}(X) \leq \lambda$.

(b) For every $\varepsilon > 0$, whenever Y is a Banach lattice, $q : Y \to X$ is a surjective lattice homomorphism, and $T : X \to Y$ is a bounded linear map with $qT = \mathrm{Id}_X$, there is a positive map $S : X \to Y$ such that

$$qS = \mathrm{Id}_X, \qquad \|S\| \leq (\lambda + \varepsilon)\|T\|.$$

Proof. Assume (a), fix $\varepsilon > 0$, and choose a positive section $U : X \to \mathrm{FBL}[X]$ with $\|U\| \leq \lambda+\varepsilon$. By the universal property, T extends to a lattice homomorphism $\widehat{T} : \mathrm{FBL}[X] \to Y$ with $\left\|\widehat{T}\right\| = \|T\|$. Both $q\widehat{T}$ and β_X are lattice homomorphisms from $\mathrm{FBL}[X]$ to X, and

$$q\widehat{T}\delta_X = qT = \mathrm{Id}_X = \beta_X \delta_X.$$

Uniqueness gives $q\widehat{T} = \beta_X$. Hence $S = \widehat{T}U$ is positive, $qS = \mathrm{Id}_X$, and $\|S\| \le (\lambda + \varepsilon)\,\|T\|$.

Conversely, apply (b), for every $\varepsilon > 0$, with $Y = \mathrm{FBL}[X]$, $q = \beta_X$, and $T = \delta_X$. This gives $\mathrm{pf}(X) \le \lambda$. □

Corollary 5.2 (Ideal formulation)**.** *A positive section for β_X exists if and only if every lattice quotient $q : Y \to X$ whose kernel is complemented as a Banach subspace has a positive right inverse. Equivalently, every such kernel is the kernel of a positive projection on Y.*

Proof. A quotient has a bounded linear right inverse exactly when its kernel is linearly complemented. A positive section S produces the positive projection $P = Sq$, with $\ker P = \ker q$. Conversely, if P is positive and $\ker P = \ker q$, then $q|_{P(Y)}$ is bijective and its inverse is positive: for $x \ge 0$, choose $y \ge 0$ with $qy = x$ and observe that $(q|_{P(Y)})^{-1}x = Py \ge 0$. □

Thus a counterexample to Question 1.1 can be sought without mentioning free Banach lattices: it is enough, and necessary, to find a linearly split lattice quotient with no positive splitting. We found no such example in the literature. Conversely, a theorem asserting that every linearly split lattice quotient splits positively would solve the problem affirmatively.

6 A universal positive lifting into the bidual

There is always a positive section after passing to the bidual. This is both a partial affirmative result and a precise description of the obstruction.

Theorem 6.1 (Canonical bidual section)**.** *For every Banach lattice X there is a positive contraction*

$$U_0 : X \longrightarrow \mathrm{FBL}[X]^{**}$$

such that

$$\beta_X^{**} U_0 = J_X,$$

where $J_X : X \to X^{**}$ is the canonical embedding.

Proof. Put $F = \mathrm{FBL}[X]$ and $I = \ker \beta_X$. Since F/I is lattice isometric to X, the adjoint

$$\beta_X^* : X^* \longrightarrow F^*$$

identifies X^* lattice isometrically with the annihilator $I^\perp$. The annihilator of a closed ideal is a band in the dual Banach lattice F^*. Dual Banach lattices are Dedekind complete, so there is a band projection

$$P_I : F^* \longrightarrow I^\perp.$$

It is positive and contractive. Define

$$R_0 = (\beta_X^*)^{-1} P_I : F^* \longrightarrow X^*.$$

Then R_0 is a positive contraction and $R_0 \beta_X^* = \mathrm{Id}_{X^*}$. Taking adjoints yields

$$\beta_X^{**} R_0^* = \mathrm{Id}_{X^{**}}.$$

The map $U_0 = R_0^* J_X$ has all the required properties. □

Proposition 6.2 (The exact weak-star criterion)**.** *There is a positive section $U : X \to \mathrm{FBL}[X]$ for β_X if and only if there is a positive weak-star-to-weak-star continuous map*

$$R : \mathrm{FBL}[X]^* \longrightarrow X^*$$

satisfying $R\beta_X^* = \mathrm{Id}_{X^*}$. In that case one can take $R = U^*$.

Proof. If U exists, U^* is positive, weak-star continuous, and $U^*\beta_X^* = \mathrm{Id}$. Conversely, a bounded weak-star-to-weak-star continuous operator R has a preadjoint $U : X \to \mathrm{FBL}[X]$. Positivity of R implies positivity of U, and taking adjoints in $R\beta_X^* = \mathrm{Id}$ gives $\beta_X U = \mathrm{Id}$. $\square$

The band retraction R_0 in Theorem 6.1 always exists. The unresolved point is that a band projection on a dual Banach lattice need not be weak-star continuous. This is not merely a technicality; the canonical R_0 behaves very badly on nonatomic examples.

For $x^* \in X^*$, let $a_{x^*} \in \mathrm{FBL}[X]^*$ be evaluation at x^* in the standard functional representation:

$$a_{x^*}(f) = f(x^*).$$

When $x^* \neq 0$, this is an atom of $\mathrm{FBL}[X]^*$.

Proposition 6.3 (Action of the canonical band projection on atoms)**.** *Let P_I be the band projection used in Theorem 6.1. Then*

$$P_I a_{x^*} = \begin{cases} a_{x^*} = \beta_X^* x^*, & \textit{if } x^* : X \to \mathbb{R} \textit{ is a lattice homomorphism,} \\ 0, & \textit{otherwise.} \end{cases}$$

Consequently,

$$U_0(x)(a_{x^*}) = \begin{cases} x^*(x), & x^* \in \mathrm{Hom}_{\mathrm{lat}}(X, \mathbb{R}), \\ 0, & x^* \notin \mathrm{Hom}_{\mathrm{lat}}(X, \mathbb{R}). \end{cases}$$

Proof. A band projection sends an atom either to itself or to zero. We have $P_I a_{x^*} = a_{x^*}$ exactly when $a_{x^*} \in I^\perp = \beta_X^*(X^*)$. If $a_{x^*} = \beta_X^* \phi$, evaluation on $\delta_X(X)$ gives $\phi = x^*$. Thus membership is equivalent to $a_{x^*} = \beta_X^* x^*$. If x^* is a lattice homomorphism, both sides are lattice homomorphisms on $\mathrm{FBL}[X]$ extending the same map on X, so uniqueness of the free extension gives equality. Conversely, if equality holds, then $x^*\beta_X$ is a lattice homomorphism; surjectivity of β_X implies that x^* itself is a lattice homomorphism. The formula for U_0 follows from the definitions. $\square$

Corollary 6.4. *If the real-valued lattice homomorphisms on X do not separate points, then for every nonzero x annihilated by all of them, the element $U_0 x$ does not belong to the canonical copy of* $\mathrm{FBL}[X]$ *inside* $\mathrm{FBL}[X]^{**}$. *In particular, this happens for every nonzero $x \in L_1[0,1]$.*

Proof. If $U_0 x = J_F f$ for $f \in F = \mathrm{FBL}[X]$, then Theorem 6.3 gives $f(x^*) = 0$ for every $x^* \in X^*$. The functional representation of $\mathrm{FBL}[X]$ is faithful, hence $f = 0$. But then $J_X x = \beta_X^{**} U_0 x = 0$, a contradiction. $\square$

This shows why the automatic bidual lifting does not settle the original problem, especially on nonatomic lattices.

7 Stability properties

The following elementary permanence results enlarge any affirmative class.

Proposition 7.1. *The positive factorization property has the following stability properties.*

(a) It is invariant under lattice isomorphism.

(b) Suppose Y has the property, $J : X \to Y$ is positive, $Q : Y \to X$ is a lattice homomorphism, and $QJ = \mathrm{Id}_X$. Then X has the property.

(c) *If each X_i has a positive section U_i with* $\sup_i \|U_i\| < \infty$, *then the* ℓ_1*-sum* $X = (\bigoplus_i X_i)_1$ *has the property.*

Proof. For (a), if $A : X \to Y$ is a lattice isomorphism and U_Y is a positive section, then

$$U_X = \mathrm{FBL}[A^{-1}]U_Y A$$

is a positive section; naturality of β for lattice homomorphisms gives the identity.

For (b), put $U_X = \mathrm{FBL}[Q]U_Y J$. Since Q is a lattice homomorphism,

$$\beta_X \,\mathrm{FBL}[Q] = Q\beta_Y,$$

and therefore $\beta_X U_X = Q\beta_Y U_Y J = QJ = \mathrm{Id}_X$.

For (c), let $j_i : X_i \to X$ be the coordinate lattice embeddings and set

$$U(x_i)_i = \sum_i \mathrm{FBL}[j_i]U_i x_i.$$

The series converges absolutely, with $\|U(x_i)\| \le (\sup_i \|U_i\|) \sum_i \|x_i\|$. It is positive, and naturality gives

$$\beta_X U(x_i) = \sum_i j_i \beta_{X_i} U_i x_i = (x_i).$$

□

Part (b) explains why lattice retracts inherit the property. The supplied Theorem 2.2 is stronger in one special direction: there Q is only positive, but the atomic coordinate formula compensates for the lack of naturality noted in Theorem 2.3.

8 Why the obvious nonatomic L_1 construction fails

The atomic formula suggests replacing a sum over atoms by an integral. This produces a bounded positive map into the ambient space of homogeneous functions, but not into the free Banach lattice.

Let $X = L_1[0,1]$, so $X^* = L_\infty[0,1]$. For $f \in L_1$ define

$$\Gamma_f(g) = \int_0^1 f(t)g(t)^+ \, dt, \qquad g \in L_\infty.$$

The map $f \mapsto \Gamma_f$ is linear and positive with respect to pointwise order on homogeneous functions. Moreover, for any finite family $g_1, \dots, g_n \in L_\infty$,

$$\sum_{j=1}^n |\Gamma_f(g_j)| \le \|f\|_1 \left\| \sum_{j=1}^n |g_j| \right\|_\infty .$$

Thus Γ_f satisfies exactly the norm estimate one would expect for an element of $\mathrm{FBL}[L_1]$.

Nevertheless, $\Gamma_1 \notin \mathrm{FBL}[L_1]$. Let (r_n) be the Rademacher functions. Then $r_n \to 0$ in $\sigma(L_\infty, L_1)$: this is immediate first against dyadic simple functions and then follows for all L_1 functions by density. But

$$\Gamma_1(r_n) = \int_0^1 r_n^+ \, dt = \frac{1}{2} \qquad (n \in \mathbb{N}).$$

Elements of $\mathrm{FBL}[E]$ are weak-star continuous on bounded subsets of E^* in the standard functional representation. Hence Γ_1 lies outside $\mathrm{FBL}[L_1]$.

There is also a finite-dimensional approximation that makes the obstruction visible. For a finite measurable partition $\mathcal{P} = \{E_1, \dots, E_m\}$ with $\mu(E_i) > 0$, define

$$U_{\mathcal{P}} f = \sum_{i=1}^{m} \left(\int_{E_i} f \, d\mu \right) \delta_{L_1} \left(\frac{\mathbf{1}_{E_i}}{\mu(E_i)} \right)^{+}.$$

Then $U_{\mathcal{P}} : L_1 \to \mathrm{FBL}[L_1]$ is a positive contraction and

$$\beta_{L_1} U_{\mathcal{P}} f = \mathbb{E}(f \mid \mathcal{P}).$$

Along refining dyadic partitions, the right-hand side converges to f in L_1. However, for $f = 1$ and $g \in L_\infty$,

$$U_{\mathcal{P}} 1(g) = \sum_i \mu(E_i) \left(\frac{1}{\mu(E_i)} \int_{E_i} g \, d\mu \right)^{+},$$

which converges pointwise to $\int g^+ = \Gamma_1(g)$. The bounded net therefore converges only in a bidual/pointwise sense, to an object outside the free Banach lattice. This is a concrete manifestation of the weak-star continuity obstruction in Theorem 6.2.

This calculation is not a counterexample: it rules out the most natural integral formula, not every possible positive section for $L_1[0, 1]$.

9 Final assessment and viable routes forward

The supplied result is mathematically sound. Its principal proof requires no repair, only two contextual corrections: the finite-dimensional case was known more strongly, and c_0 is already affirmative. The exact scope of the positive-retract hypothesis is the class of atomic L_1 spaces $\ell_1(A)$.

The strongest additional full theorem located is the AM-space consequence of Lotz's positive lifting theory, which yields all $C_0(K)$ and all $c_0(\Gamma)$. The general problem, including the basic nonatomic test case $L_1[0, 1]$, remains unresolved in the literature we found.

The reductions above suggest two focused approaches.

(1) By Theorem 5.1, construct a Banach lattice Y and a linearly complemented ideal I such that the quotient map $Y \to Y/I$ has no positive right inverse. This would give a counterexample immediately.

(2) By Theorem 6.2, seek a positive weak-star continuous retraction of $\mathrm{FBL}[X]^*$ onto the band $\beta_X^*(X^*)$. The canonical band projection is positive and contractive but is generally not weak-star continuous; any successful proof must replace it by a different positive retraction or prove additional structure forcing weak-star continuity.

The finite-partition computation for L_1 shows that a bare compactness or local finite-dimensional argument is unlikely to suffice: it naturally produces the canonical bidual section rather than an element of $\mathrm{FBL}[L_1]$.

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